STOPS AND STOPPING REGIMES

The frequency of transit unit stopping along a line, or distance between the locations where the TU stops for passengers, affects both passenger travel time and efficiency of operations. It is therefore important to plan stop locations and stopping regimes carefully for each line.

- **Stopping Regime** – it represents the method of selecting stoppings, i.e. determination whether Tus make stoppings at all designated stops or only at all designated stops or only at those at which passengers request.
- **Stops** are locations designated for passenger pick-up and drop-off by TUs. Depending on the type or regime of operation, stops may be designated as demand, on-call or fixed.

- **Demand Stopping** is the regime when TUs stop at any location that passengers on board or along a street demand
- **On-call stopping** is the operating practice in which there are fixed stop locations, but TUs stop only at stops where passengers demand
- **Fixed Stopping** is the regime in which all TUs stop at all stopping locations - stops or stations – regardless of whether any passengers will board or alight

ACCELERATED OPERATIONS WITH FIXED STOPPING SCHEDULES

- If station locations are fixed, the only way to increase speed is to introduce **accelerated operations**
- Basic accelerated operations are **skip-stop and zonal operation**

Skip-Stop Operation

- Skip stop is the only type of operation by which operating speed on transit lines with only two tracks can be increased while maintaining high frequency and high line capacity
- TUs are operated in two patterns: bus A stop at A and AB stations, while bus B stops at B and AB stations. A and B stations are served by every other TU, while at AB stations all TUs stop
- Skip-stop operation has higher operating speed and fewer stoppings along the line. Passenger travel time is reduced and service is more attractive
- For a constant no. of passengers, total transportation network (prs-km) of the line remains the same. However, increased operating, and consequently cycle speed of TUs results in their higher productivity (prs-km/veh-h). If this allows a decrease in the no. of TUs, there are substantial cost savings

Methodology for Skip-Stop Planning and Evaluation

Data Collection – This calls for obtaining operating data of the analysed line : L, T_o, h, C_v etc.

Planning decisions - The alternative skip-stop combinations that are to be analysed

Data Preparation – The trip table is prepared. This table includes all required passenger volumes

Performance Computation – Passengers saving time is calculated

Evaluation of Alternatives -

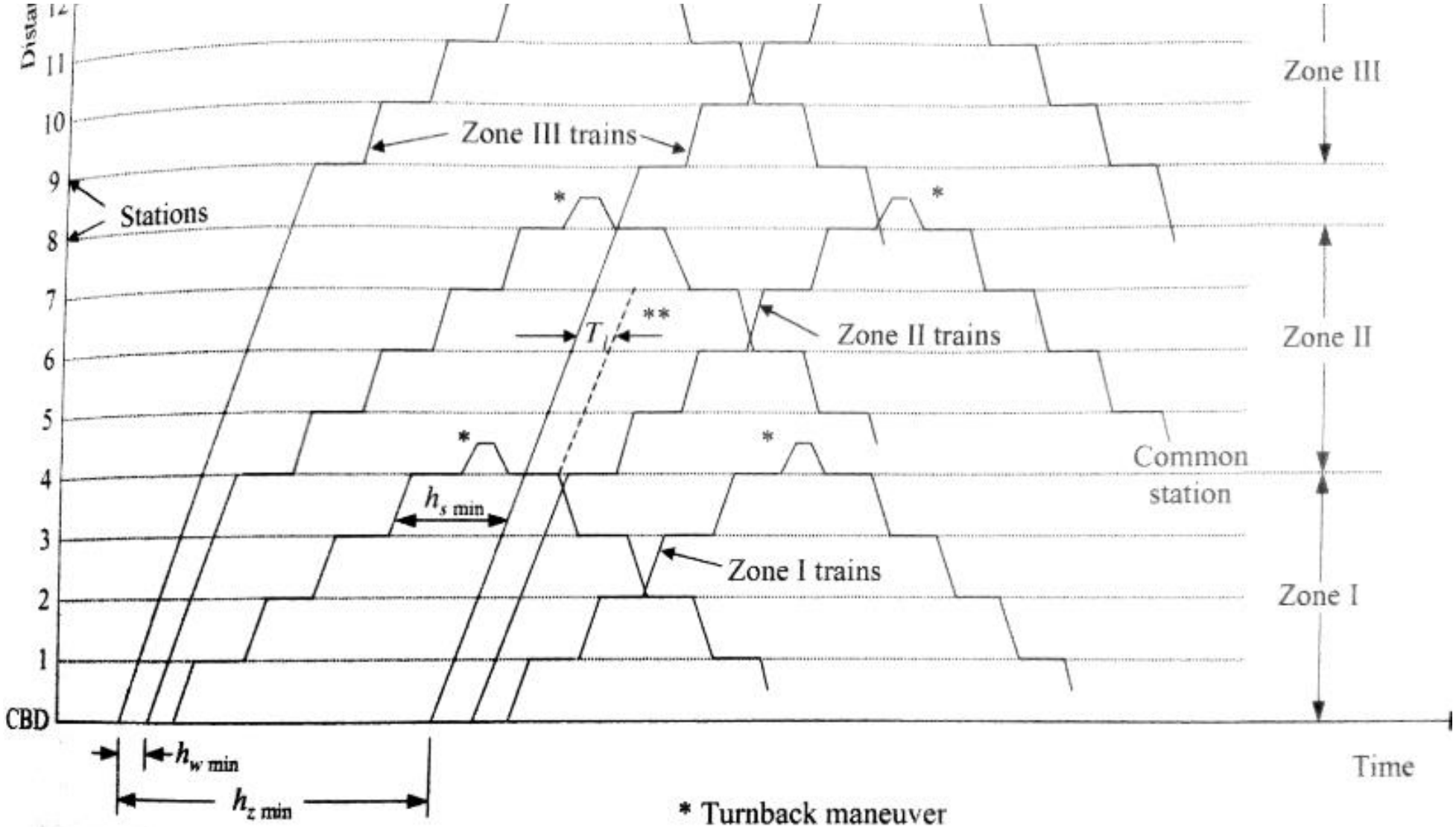
ZONAL OPERATION

The operating pattern of trains with respect to zone terminals can be of several types:

- There are no joint stations among zones, such as between zones II and III. All stations have the same service frequency, and travel between the two zones is practically impossible

- Trains of both adjacent zones serve the terminal station of each zone, as between zones I and II. These stations have double frequency, and travel between adjacent zones is possible through one transfer between consecutive trains

- Each zone terminal is also served by trains serving all zones beyond it



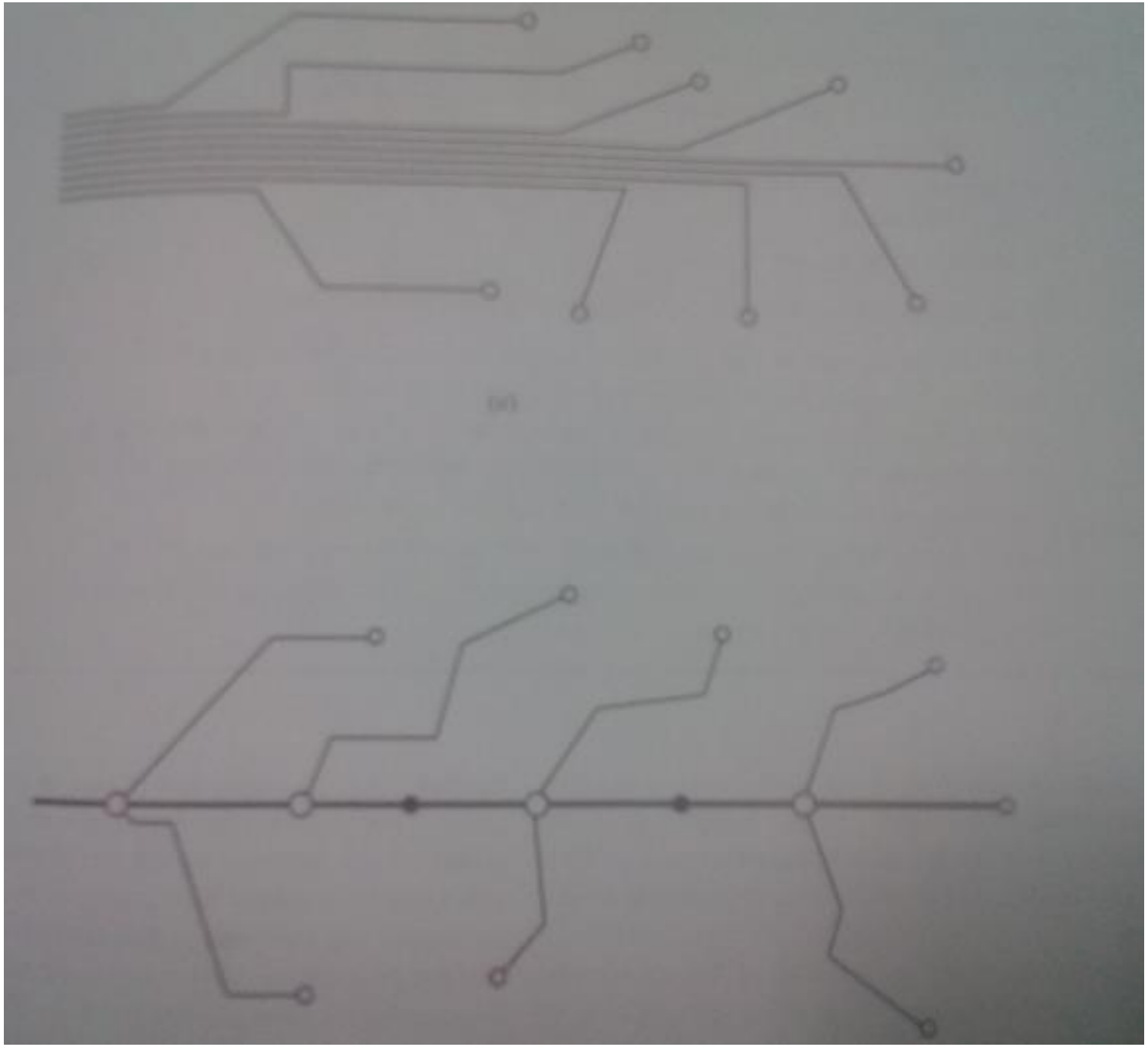
* Turnback maneuver

** Option with Zone III train stopping at station 4: zone II and zone I trains would have to be delayed by T_l

***Figure 2.17** Time-distance diagram for zonal operation.*

TYPES OF TRANSIT LINES AND THEIR CHARACTERISTICS

- Radial
- Diametrical
- Tangential
- Circle or ring
- Trunk with branches
- Trunk with feeders



Branches, compared to feeders, have the following advantages:

- Provide for passengers a continuous service without transfers and delays between center city and suburban areas
- Being longer lines, trunk and branches have smaller percentage of cycle time as terminal times than trunk with feeders
- Transfer stations are not needed

Feeders, on the other hand, have the following advantages over branches:

- Each line section can be optimized with respect to mode and vehicle type (or TU) used, schedule etc.; load factors are higher, fleet size smaller, and operating costs lower
- Use of higher performance vehicles or mode (usually rail) on the trunk provides superior service at lower operating cost than smaller vehicles from branches (buses) can provide
- Regular headways can be operated on the trunk and on each feeder line
- More reliable service: delays are less likely to transfer between trunk and feeders
- Suburban terminals for trunk offer not only trunk/feeder transfers, but also transfers among feeder lines, providing greater network connectivity

TRANSFERS IN TRANSIT NETWORKS

- To attract passengers and compete with private automobiles throughout an urban area, transit must offer integrated service by area-wide networks
- To achieve intramodal integration of different lines (e.g. within metro or bus networks), as well as intermodal integration, convenient and efficient transfers among lines must be provided
- Passenger transfers among transit lines and modes involve a certain resistance because they interrupt travel and require passengers to orient themselves and walk between vehicles on different lines; they also cause some delay
- When transfers are planned correctly and resistance is minimized through good station design, convenient walking paths, station amenities, schedule coordination etc.
- The inconvenience to passengers can be so small that it is easily outweighed by the benefits transfers bring with respect to the availability of different travel routings through the network, shorter headways on some lines, and generally better services

Integration of Public Transportation Modes

Objectives of thisSession

- Know the types of Intermediate Public Transport operating around the globe
- Know the advantages and disadvantages of IPT
- Learn the operations and routing of IPT
- Know the Policy Issues in managing IPT in India

Intermediate Public Transport

- Refers to modes which bridge the gap between private transport and formal public transport modes in cities
- Two broad categories are
    - Taxi /auto-rickshaw services
    - Informal public transport

Global Presence of IPT

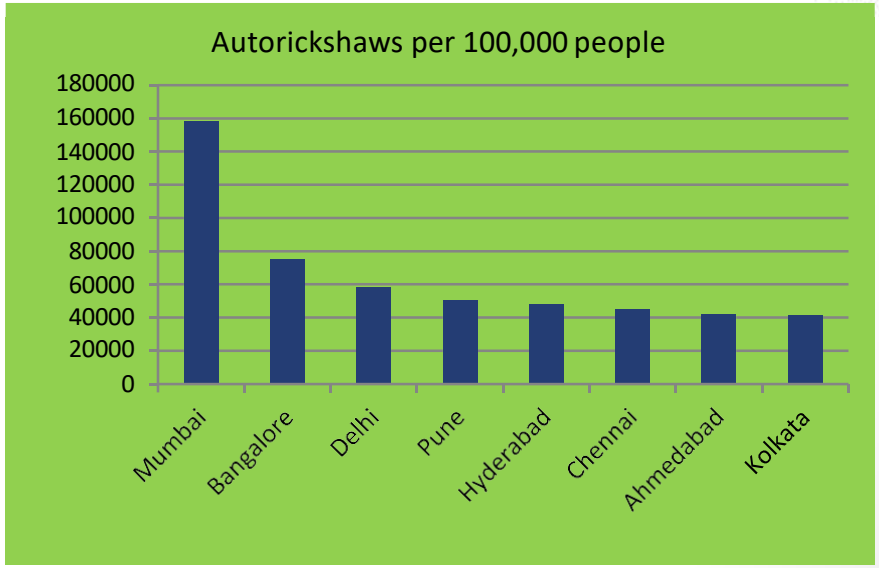
- Principal means of IPT can be found in most Developing and Underdeveloped countries of the world
- Dominant in most of the cities in Asia, South Africa, and South America.

An overloaded auto rickshaw in Pakistan

IPT in Indian Cities

- Large metropolitan cities having organized PT systems exhibit wide coverage of auto-oriented modes
- Medium and smaller sized cities exhibit a wide range of manually-operated para transit modes.

- Major metro cities exhibit a higher share of auto rickshaws, ranging between 7 – 13 auto rickshaws per 1000 people
- Smaller cities supply ranging between 0.3 to 2 auto rickshaws per 1000 people



Advantages to IPT

Connectivity to Public Transport - Providing connectivity, among other parameters such as frequency, reliability, comfort, and safety

Door-to-Door Service as opposed to Private Motorized Transport - Ability to make door-to-door (long-distance) trips is a clear benefit over private motor vehicles.

Source of Employment - For thousands of unskilled and less educated sections of society the world over

Safety – Lower speeds and light weight, IPT modes are relatively safer in operation

Operations & Routing

The permits issued to an auto rickshaw driver are generally of two types:

- Route permits - Route permits are permits issued to auto rickshaws for operating on a particular route
- Zone permits - Zone permits are based on permits issued to an auto rickshaw for plying in a particular zone of the urban area

Management Options of IPT

- An agency responsible for the provision of IPT has the option of managing the system itself or contracting for one or more functions

IPT practitioners would benefit from:

A national database that has expanded information relating to organizational and management structure

Guidance on the criteria used to determine the appropriate organizational and management structure

Policy Issues for IPT

- Need for zone-wide permits, rather than city-wide
- ITS applications for monitoring the operations of Autos/Taxis
- Rational fare fixation policy to be evolved in tune with input costs
- Streamlining the procedures for infrastructure facilities
- Designing organization structure for monitoring operations
- Public information about IPT system
- Integration of IPT with other modes
- Making IPT business commercially viable and prescribing entry restrictions

Integration With Other Modes

- The National Urban Transport Policy (NUTP) envisions auto-rickshaw (and other types of contract carriage) services as modes that serve "occasional trips such as trips to airports or rail stations with excessive baggage, or emergency trips that have to be undertaken immediately when it is not possible to wait for public transport"
- IPT should provide the first and last mile connectivity to public transport
- IPT should act as a feeder to public transport services while ensuring accessibility to commuters with special needs, such as elderly and persons with disabilities

Image Makeover

- Commuters are generally dissatisfied with the services provided by IPT in India
- IPT lacks in aspects of traffic safety, personal security, and service image.
- Commuter's attitudes towards IPT will be an important key for future development

Auto rickshaw stand in front of Central Railway Station in Chennai, India

Image Makeover cont'd…

- Banning auto rickshaws contributes to rising tide of car dependence
- On the other hand, an improved auto rickshaw sector can be a key element of a transport system that presents a viable alternative to driving.
- Auto rickshaws should become less polluting by running on CNG – Delhi's pollution level was reduced by doing so

Right Enforcement

- Safety and quality of service of auto rickshaws can be addressed through tighter regulation and the use of technology.
- Delhi traffic police have already announced that they will verify auto rickshaw drivers' credentials, issue badges and biometric IDs, and create a database of drivers' addresses. This will reduce unlicensed operation.

Service Restructuring

A coordinated effort to make the business safer, healthier, and more financially secure for drivers would reduce accidents and improve customer service.

Include initiatives such as:

- Driver education
- Regulated working hours
- Stricter emissions standards
- Micro-financing opportunities
- Insurance policies

A scene in Hyderabad, India

Deficiencies in IPT

- Auto stands are not located in many areas/localities, and hence there is a lack of suitable facilities
- Auto Rickshaw operations are not monitored by regulating agencies
- There is no credibility for behavior of Auto drivers
- There is no accountability by Auto owners
- There is no security for women, especially in late night operations
- In many autos, fare meters are tampered
- Many Auto drivers “refuse to come” when passengers want to go to a particular place
- There are many Autos operating without proper permits (e.g. Duplicate permits)
- There are many old vehicles which need to be replaced.